```
PS C:\Users\nisch\Desktop\EECS658\NischayRawal_Assignment5> python .\Assignment5.py

==========================================================================
Part 1: Imbalanced Data Set
==========================================================================
Confusion Matrix:
Class Order: ['Iris-setosa', 'Iris-versicolor', 'Iris-virginica']
[[39  1  0]
 [ 0 25  5]
 [ 0  2 48]]
Accuracy: 0.933333
Class Balanced Accuracy: 0.904665
Balanced Accuracy: 0.943929
balanced_accuracy_score (scikit-learn): 0.922778

==========================================================================
Part 2: Oversampling
==========================================================================

==========================================================================
Random Oversampling
==========================================================================
Confusion Matrix:
Class Order: ['Iris-setosa', 'Iris-versicolor', 'Iris-virginica']
[[39  1  0]
 [ 0 26  4]
 [ 0  3 47]]
Accuracy: 0.933333
```

==========

SMOTE Oversampling

==========

Confusion Matrix:

Class Order: ['Iris-setosa', 'Iris-versicolor', 'Iris-virginica']

```
[[39  1  0]
 [ 0 25  5]
 [ 0  4 46]]
```

Accuracy: 0.916667

==========

ADASYN Oversampling

==========

Confusion Matrix:

Class Order: ['Iris-setosa', 'Iris-versicolor', 'Iris-virginica']

```
[[39  1  0]
 [ 0 26  4]
 [ 0  2 48]]
```

Accuracy: 0.941667

==========

Part 3: Undersampling

==========

==========

Random Undersampling

==========

Confusion Matrix:

Class Order: ['Iris-setosa', 'Iris-versicolor', 'Iris-virginica']

```
[[39  1  0]
 [ 0 26  4]
 [ 0  4 46]]
Accuracy: 0.925000

======================================================================
Cluster Centroids Undersampling
======================================================================
Confusion Matrix:
Class Order: ['Iris-setosa', 'Iris-versicolor', 'Iris-virginica']
[[39  1  0]
 [ 0 26  4]
 [ 0  3 47]]
Accuracy: 0.933333

======================================================================
Tomek Links Undersampling
======================================================================
Confusion Matrix:
Class Order: ['Iris-setosa', 'Iris-versicolor', 'Iris-virginica']
[[39  1  0]
 [ 0 25  5]
 [ 0  2 48]]
Accuracy: 0.933333
PS C:\Users\nisch\Desktop\EECS658\NischayRawal_Assignment5>
```